

Yosys was put to test with the designs of OpenRISC 1200 CPU, openMSP430 CPU, OpenCores I2C master and k68 CPU

techniques. ABC then maps the internal gate library of Yosys to the target architecture. To validate the logic formula in the design, Yosys takes help from MiniSat, the SAT-solver library.

The Yosys flows

You are free to choose the best format for your design and allow data to flow from input to output in a sequence. The main difference between these flows would be the mapping to library cells. A straight-forward application-specific integrated circuit (ASIC) design is mapped to simple CMOS cells. When synthesising for field programmable gate array (FPGA) based designs, the design has to be mapped to look-up tables (LUTs) that are stored in the library.

Yosys interconnects to Xilinx library to allow for your FPGA to be designed. For logic mapping to digital signal processing (DSP) cells in FPGAs, or synthesis for custom coarse-grain reconfigurable hardware, you can choose to go with the coarse-grain architecture, whose cells operate on bit-vectors, instead of single-bit signals, and accelerate the process.

Check for correctness

As soon as you install your tool, check if it is functioning right. Yosys-bigsim on GitHub provides a vari-

Want to try Yosys but do not know Verilog? Simply log on to verilog.james.walms.co.uk. The blog "Learn Verilog with Yosys" by James Walsmley and hosted by GitHub shows you the way.

You can even directly get on it with the following instructions:

```
~ $ yosys
# Open your imagination
yosys> read_verilog fiedler-cooley.v
# Learn verilog!
```

ety of real-world Verilog designs and test benches. Bigsim compares the results of simulation, before and after synthesis, to tell you if your design is still what you intended it to be.

To check the correctness of the Verilog front-end of Yosys, a Verilog regression tester called Vlog-Hammer is used. This currently tests combinatorial circuits described by Verilog expressions by comparing the results from Yosys with other synthesis-tool results, while at the same time also cross-verifying using various simulation engines.

Coming to the verification of your design, a symbolic model check (SMC) formally checks the properties of your circuit. To check if two circuits function identically, you may reproduce a circuit, or you may check the design at two different stages of tool processing, to perform a formal equivalence check. You can also

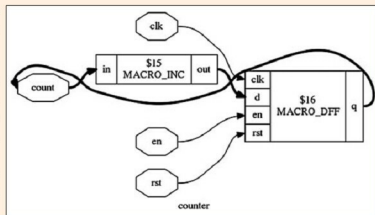


Fig. 3: Coarse-grain netlist

check if a module conforms to interface standards.

What more would you like

Using Emscripten, an exclusive JavaScript port is being developed for Yosys. Called YosysJS, it is meant for building Web applications for educational purposes. When you contrast Yosys with proprietary tools, the main drawback seems to be advanced-level synthesis for timing, power and such factors. May be future releases could see a tweak in Yosys that would cater to these features and much more. Stay tuned to its website www.clifford.at/yosys for regular updates. ●

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